

DATABASE PROJECT REPORT

Gym Membership & Personal Training Planner
(Project ID: #9)

A. PROJECT IDENTITY

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Course	INT1313 - Database Basics
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Table of Contents

B. REPORT STRUCTURE

1. Introduction & Project Scope (ISO/IEC/IEEE 29148 Standard)

1.1. System Objective

- (Content) In the modern fitness industry, managing memberships, tracking personal training schedules, and maintaining equipment manually often lead to data errors, scheduling conflicts, and operational inefficiency.
- (Overall Objective) To address these challenges, the Gym Membership and Personal Training Management System is designed to digitalize and streamline core gym operations. (Specific Objectives) Through a centralized relational database, the system efficiently manages member profiles, class schedules, personal trainer bookings, and fitness facility maintenance records. It aims to ensure strict data consistency and integrity, prevent scheduling conflicts, and maintain accurate subscription and billing records to support reliable data tracking and efficient gym management.

1.1.1. Core Entity Definitions & Business Semantics

To establish a common understanding of the system and avoid ambiguity in the requirements, the following key terms are defined:

- **Member:** Refers to an individual who is registered in the gym's system and can use the gym's services. A Member may remain in the system indefinitely, even when they do not currently have an active Membership. Core attributes include **Member_ID** (Primary Identifier), **Full_Name**, a unique **Phone_Number**, Profile pictures or verification photos.
- **Membership:** A transactional entity representing an active or historical contract linking a registered Member to a specific Membership Plan over a defined duration (from **Start_Date** to **End_Date**). It tracks the specific lifecycle Status (e.g., 'Active', 'Expired', 'Cancelled'). A Member can hold zero or multiple Membership records over their lifetime, but at any single timestamp, a member can have at most one active Membership. When a package expires or is terminated, the record is permanently retained in the database as historical information rather than deleted.
- **Membership Plan:** Defines the standardized membership packages pre-set by the gym (e.g., Monthly, Annual, VIP). It includes **Plan_ID** (Primary Identifier), **Plan_Name** (UNIQUE), **Duration_Months** (with constraint CHECK (Duration_Months > 0)), and **Price** (NOT NULL, with constraint CHECK (Price > 0)).

- **Daily Pass Ticket:** A transactional entity representing single-visit daily passes purchased by non-registered walk-in guests. It tracks basic guest contact details (Guest_Name, Phone), the Visit_Date, and the Daily_Rate paid, serving as an auditing log for front-desk cash receipts. Each ticket is uniquely identified by Ticket_ID.
- **Billing Log:** An automated financial ledger tracking all cash inflows. It supports both subscription billings (linked to a registered Customer via Sub_ID) and walk-in ticket sales (linked to a Daily_Pass_Ticket via Ticket_ID). It tracks Billing_Status ('Pending', 'Paid', 'Failed'), Invoice_Date, Due_Date, and the Payment_Date (set upon payment), providing a comprehensive audit trail and supporting automated billing workflows.
- **Session Equipment Usage:** An associative relationship entity representing the many-to-many (M:N) relationship where gym sessions (group Class or private PT Booking) reserve physical Equipment. It tracks specific Equipment_ID reservations for a Class_ID or Booking_ID, ensuring machines are reserved dynamically.
- **Staff:** The superclass entity representing all employees working at the gym. It is identified by Staff_ID (Primary Identifier) and tracks Name, Role ('Trainer', 'Receptionist', 'Manager'), and Monthly Salary (with constraint CHECK (Salary > 0)).
- **Trainer / PT:** A specialized subclass of Staff representing Personal Trainers. They possess unique attributes such as Specialty (e.g., Yoga, Strength) and Certification. Trainers are directly responsible for leading public group classes or private one-on-one PT bookings.
- **Receptionist:** A specialized subclass of Staff responsible for managing front-desk shifts (Shift: Morning, Evening, Night) and signing off on hardware maintenance logs.
- **Room:** Manages the physical infrastructure and training spaces of the gym. Identified by Room_ID (Primary Identifier), it tracks Room_Name (UNIQUE), Room Type (e.g., 'Group Studio', 'Open Gym Floor', 'Pilates Room'), and physical Max Capacity (with constraint CHECK (Capacity > 0)).
- **Class:** Represents public group classes (Zumba, Yoga, HIIT) scheduled by the gym. Attributes include Class_ID (Primary Identifier), Class_Name, Room_ID (physical room), and Trainer_ID (instructor). It is scheduled with precise Start_Time and End_Time timestamps, utilizing dynamic validation to prevent room and trainer scheduling overlaps.
- **Class Registration:** An associative entity representing the many-to-many (M:N) relationship where a registered Customer registers for a group Class. It uses a composite primary identifying attribute {Class_ID, Customer_ID}. Registrations are restricted to customers with active subscriptions.
- **PT Booking:** Manages private one-on-one sessions between a registered Customer and an assigned Personal Trainer (PT). Identified by Booking_ID (Primary Identifier), it is scheduled in a physical Room_ID with precise Start_Time and End_Time timestamps, subject to strict overlapping interval constraints.
- **Equipment:** Manages physical workout machines and assets. Identified by Equipment_ID (Primary Identifier), it tracks Name, Type (Cardio, Strength, Recovery), and operational Status ('Active', 'Under Maintenance', 'Broken'). If equipment status is broken, the system blocks bookings involving it.
- **Maintenance Log:** A weak entity representing the history of service and repairs on a specific equipment. Because it cannot exist without its parent machine, it uses a

composite primary identifying attribute {Equipment_ID, Log_no} (where Log_no is a partial key). It tracks Maintenance_Date, Cost (CHECK (Cost >= 0)), Description, and Logged_By (receptionist/manager signing off on the repair).

1.1.2. Schedule & Resource Conflicts

- **Schedule and Classification:**
A schedule is a systematic allocation of gym resources and time. The system classifies schedules into two main types:
 - Centralized/Public Gym Schedule: Fixed timeentity sets for group sessions (such as Yoga, Zumba, HIIT) managed and published by the gym. Members do not create private centralized schedules; instead, they register for these public classes, and the system records their attendance.
 - Personal Trainer Schedule: The private working timeentity sets and coaching slots of individual PTs for one-on-one private sessions.
- **Schedule Components:** A complete schedule record is composed of three mandatory business components: Workout Location / Equipment (e.g., Studio A, Cardio Area, Running Machine #5), Time Interval (defined by Start_Time and End_Time timestamps), and the assigned Personal Trainer (PT) coaching at that location during that interval.
- **Schedule Conflict:** Defined as an error or resource collision state in the system when more than one session is scheduled at:
 - The same Workout Location during overlapping time intervals (e.g., Studio A is assigned to two different classes simultaneously).
 - OR the same Personal Trainer (PT) during overlapping time intervals (causing a double-booking issue).
 - Domain Scope: This definition focuses strictly on resolving physical resource (location) and direct personnel (PT) scheduling conflicts during fitness activities, excluding broader administrative HR management.
- *Design Note on Scheduling Architecture: We selected a two-table model (CLASS and PT_BOOKING) to maintain a clean business separation between public group classes and private personal training sessions. We are fully aware that this architectural choice requires cross-table dynamic validation (to be enforced via Database Triggers in Phase 2) to prevent room and trainer overlaps. An alternative design approach would be merging these two tables into a single unified ScheduledEvent entity, which would simplify schedule conflict constraints at the database level but merge distinct business logics.*

1.2. Business Rules & Constraints

1.2.1. Membership & Subscription Domain (Triệu Quang)

Four object: Member, Membership_Plan, Subscription, Recurring Billing/Billing_Record

- BR-M01: Each member must have a unique member identifier (Member_ID).
- BR-M02: Each membership plan must have a unique plan identifier (Plan_ID), a defined positive price, and a valid duration.

- BR-M03: Each membership belongs to exactly one member and is associated with exactly one membership plan.
- BR-M04: Each membership record must have a defined start date (Start_Date), end date (End_Date), and lifecycle status (Status).
- BR-M05: The status of a membership must be strictly one of the following values: 'Active', 'Expired', or 'Cancelled'.
- BR-M06: An active subscription can generate multiple billing records according to its defined billing cycle.
- BR-M07: Each billing record must be associated with either exactly one membership (Membership_ID) or exactly one daily pass ticket (Ticket_ID), but never both (Exclusive OR / XOR constraint). (Membership_ID # Ticket ID)
- BR-M08: Billing records must be retained as historical financial records and must not be deleted when the associated subscription expires or is cancelled.
- BR-M09: Each billing record must store the amount charged (Amount_Paid), billing status (Billing_Status), invoice date, and payment date.
- BR-M10: At any single timestamp, a member may have at most one active membership (Status = 'Active').

1.2.2. Training & Scheduling Domain (Trần Đức Phương)

1. Problem Survey Questions

To identify domain constraints and establish proper scheduling rules, the system requirements were analyzed using the following targeted survey questions:

- **For Trainers:**
 - *Have you ever experienced double-booking conflicts between a group class and a 1-on-1 PT session in your personal schedule?*
 - *When a member cancels a private session at short notice, how does the current workflow handle fill-ins or schedule updates?*
- **For Gym Management:**
 - *How often do resource collisions (overlapping usage of trainers, members, or physical rooms) cause operational issues or customer complaints?*
 - *What are the primary challenges in tracking equipment maintenance logs and linking facility availability to class schedules?*

2. Domain Module Overview

- **Scope:** Manages group class schedules, private 1-on-1 personal training (PT) bookings, trainer assignments, facility room allocations, and session-level equipment usage.
- **Core Entities & Relationships:** STAFF/TRAINER, CLASS, PT_BOOKING, MEMBER (CUSTOMER), ROOM, and SESSION_EQUIPMENT_USE.
- **Primary Objective:** Maintain schedule integrity, optimize room capacity, and automatically enforce **Conflict Detection** across all human resources and physical facilities.

3. Business Rules (BR)

Business Rules (BR) are concise, clear, and precise statements that describe the policies, regulations, conditions, or operational constraints of an organization or enterprise.

Trainer:

- **BR-TR-01:** A trainer can teach multiple group classes (LEADS relationship).
- **BR-TR-02:** A trainer can conduct multiple 1-on-1 personal training sessions (COACHES relationship).
- **BR-TR-03:** A trainer cannot teach two classes or PT sessions at the same time.

- **BR-TR-04:** If a trainer already has a scheduled class or PT booking during a specific time period, they cannot accept another session whose start or end time overlaps with the existing scheduled session (e.g., if a trainer is scheduled from 09:00–10:00, they cannot accept a session from 09:30–10:30).

Class (Group Training Class):

- **BR-CL-01:** A class must have exactly one trainer (LEADS).
- **BR-CL-02:** A class must have a specific Start_Time and End_Time.
- **BR-CL-03:** The Start_Time must be earlier than the End_Time (Start_Time < End_Time).
- **BR-CL-04:** A class must have a defined maximum capacity (Max_Capacity).
- **BR-CL-05:** A full class cannot accept additional registrations (REGISTER / CLASS_REG).
- **BR-CL-06:** A cancelled class cannot accept new registrations.
- **BR-CL-07:** A class must be hosted in exactly one room (HOSTS_CLASS).
- **BR-CL-08:** A class can utilize equipment during a session via SESSION_EQUIPMENT_USE (CLASS_USES - Arc XOR constraint).

PT_Booking (Personal Training Session):

- **BR-PT-01:** A PT booking represents a 1-on-1 session conducted by exactly one trainer (COACHES) and booked by exactly one member (BOOKS_PT).
- **BR-PT-02:** A PT booking must have a specific Start_Time and End_Time, where Start_Time < End_Time.
- **BR-PT-03:** A PT booking has a Status (e.g., Confirmed, Cancelled, Completed).
- **BR-PT-04:** A member cannot book a PT session that has already started or completed.
- **BR-PT-05:** A PT booking must be hosted in exactly one room (HOSTS_PT).
- **BR-PT-06:** A PT booking can reserve/use equipment during a session via SESSION_EQUIPMENT_USE (PT_USES - Arc XOR constraint).
- **BR-PT-07:** An equipment usage instance (SESSION_EQUIPMENT_USE) can be associated with either a Group Class (CLASS_USES) OR a PT Booking (PT_USES), but mutually exclusive (Arc XOR constraint).

4. Scheduling Conflict Detection

Conflict is defined as an error or resource collision state within the system that occurs when more than one session (group class or private PT booking) is scheduled during overlapping time intervals.

- **1. Trainer Conflict:**
 - A trainer cannot teach two or more sessions (Classes or PT Bookings) at the same time.
 - *Example:* If Trainer T01 is assigned to Class 1 (09:00–10:30) and Class 2 (09:30–10:30), these two sessions overlap from 09:30 to 10:30 → **Conflict: Not Allowed.** The system must reject the second assignment.
 - *Boundary Case:* If Session 1 ends at 10:30 and Session 2 starts at 10:30 → **No Conflict: Allowed.**
- **2. Member Conflict:**
 - A member cannot participate in multiple sessions (Classes or PT Bookings) simultaneously.
 - *Example:* If Member M01 attempts to book PT Session 1 (09:00–10:00) and Class 2 (09:30–10:30), the time overlap (09:30–10:00) causes a collision → **Conflict: Not Allowed.** The system must reject the second booking.
- **3. Room Conflict:**
 - A room cannot host more than one session (Class or PT Booking) at the exact same time (HOSTS_CLASS vs HOSTS_PT).
 - *Example:* If Room R01 hosts Class A from 14:00–15:00, no PT booking can be assigned to Room R01 during 14:00–15:00.

1.2.3. Facility & Maintenance Domain (Nguyễn Văn Hoàng Te)

- **Rule 1 (Asset Identification - BR-FAC-01):** Fitness equipment (Equipment) is a strong entity, uniquely identified by Equipment_ID. Each equipment tracks Name, Type (Cardio, Strength, Recovery), and operational Status (Active, Under Maintenance, Broken).

Business & Technical Explanation: This rule governs gym assets. Every high-value fitness machine is modeled as a strong entity uniquely identified by its equipment_ID. To support sorting and scheduling, the system tracks: Name, Type (which must be classified as Cardio, Strength, or Recovery), and Status (which tracks whether the machine is Active, Under Maintenance, or Broken) which serves as the scheduling availability flag.

- **Rule 2 (Weak Entity Constraint - BR-FAC-02):** A maintenance log (Maintenance_Log) cannot exist without a parent equipment. It is modeled as a weak entity with a partial key Log_no (representing the sequential maintenance attempt for a specific machine, e.g., attempt 1, 2, 3).

Business & Technical Explanation: Maintenance logs are sequential events that have no meaning or existence without being linked to a specific machine. According to database design theory, it is modeled as a Weak Entity (represented by a double-bordered rectangle in the EER diagram). Its partial key is Log_no, representing the sequential service number for that specific machine (e.g., Treadmill #5 has Log_no 1, 2, 3...). This avoids independent ID bloat and reflects structural dependency.

- **Rule 3 (Composite Primary Identifier - BR-FAC-03):** The identifying key of Maintenance_Log is a composite key combining the owner's key and the partial key: {Equipment_ID, Log_no}. This allows the same machine to undergo multiple repairs over time without identification conflicts.

Business & Technical Explanation: The identifying key of the weak entity is a composite identifier: {Equipment_ID, Log_no}. This uniquely identifies each maintenance log, allowing the same machine (fixed Equipment_ID) to undergo unlimited maintenance sessions over time as Log_no increments (1, 2, 3...) without duplication, preserving a complete lifecycle log of the asset.

- **Rule 4 (Participation & Dependency - BR-FAC-04):** The relationship LOGGED_FOR between Equipment and Maintenance_Log has a total participation constraint (double line) on the Maintenance_Log side (every log must belong to a machine), and a partial participation constraint (single line) on the Equipment side (new machines may have zero maintenance logs).

Business & Technical Explanation: This rule defines the EER participation constraints between Equipment and Maintenance Logs:

- Maintenance_Log side (Double Line - Total Participation): Mandates that every maintenance record must refer to an existing machine. "Orphan" service expense logs without a machine ID are strictly blocked to ensure transparent financial auditing.

- Equipment side (Single Line - Partial Participation): A newly purchased, functional machine does not require any maintenance logs, providing operational flexibility.

- **Rule 5 (Staff Sign-off Chain - BR-FAC-05):** *Every maintenance log must be signed off and performed by exactly one Staff member (specifically a Receptionist or Manager) via the PERFORMS_MAINTENANCE relationship. This establishes a strict responsibility audit trail.*

Business & Technical Explanation: This establishes administrative accountability for gym operations. Repairs and servicing cannot occur unmonitored; they must be supervised and signed off by a receptionist or manager. This audit trail ensures that if a machine fails and causes user injury shortly after being serviced, the manager can identify the employee who approved the maintenance. This role-limiting check represents an operational rule that will be validated by the system during maintenance logging

- **Rule 6 (Operational Dependency - BR-FAC-06):** If a machine's status is changed to 'Broken' or 'Under Maintenance', the system blocks any new session reservation utilizing this equipment and flags any existing future reservations to prompt equipment reassignment.

Business & Technical Explanation: This rule ensures operational synchronization. When a technician marks an equipment's status as 'Broken' or 'Under Maintenance', the database automatically triggers a validation rule. It scans future session reservations in the SESSION_EQUIPMENT_USAGE relationship, raises alert notifications for cancellations or reassignments, and blocks new schedules from reserving it, preventing physical scheduling conflicts.

- **Rule 7 (Financial Validation - BR-FAC-07):** *The maintenance Cost recorded in each log must be a non-negative value (Cost ≥ 0.00).*

Business & Technical Explanation: This rule validates financial data inputs. The maintenance Cost stored in the log must be non-negative (defaulting to zero if routine service is free under warranty). This prevents typographical errors where negative numbers could distort the gym's expense reports and financial audits.

2. Database Design (Conceptual ER/EER Modeling)

